

grna assignment results

2026-01-10

outline

- methods
- datasets (real and simulated)
- results: simulation ground truth analysis and real data

1. Methods and datasets

For this part of the project, we are assessing five methods (two of which are versions of `sceptre`) on two real datasets and one simulated dataset.

The methods:

1. `pertpy`'s `assign_mixture_model()`: Bayesian Poisson-Gaussian mixture model fit via MCMC with CUDA+JAX support
2. `crispat`'s `ga_poisson_gauss()`: a Poisson-Gaussian mixture fit where the MAP estimates are obtained from a variational procedure
3. `cleanser`: Bayesian NB-NB mixture or Pois-NB mixture fit via MCMC with `stan` (their `direct-capture` and `crop-seq` methods, respectively)
4. `sceptre`'s mixture model: latent variable Poisson GLM fit via EM variant
5. new `sceptre`-pipeline

The datasets:

- a. Gasperini 2019 (high moi)
- b. Replogle 2022 (low moi)
- c. simulations

1a. Simulated data

Simulated data: we want to try these out on simulated data where we know the ground truth.

Approach: we wanted to include covariate effects. To get these, we ran `sceptre-pipeline` on the Replogle-rd7 data, and restricted to guides with the total number of assigned cells between the 95% and 99% percentile. We then picked the guide that had the most non-zero UMI counts. (TODO is this not ideal? maybe something odd happened to that guide to make it that extreme.) Our simulation is based on this guide, call it g .

To learn the covariate effects on the expressions, we did a NB regression of the UMI counts for g on our covariates ($\log(\text{grna_n_nonzero}+1)$ and $\log(\text{grna_n_umis}+1)$, both of which were significant). That lets us model the marginal mean (marginal over the mixture), but also the likelihood is dominated by the non-perturbed cells, so it's mostly just the covariate effect for those cells. This model gives us $\hat{\beta}$ and $\hat{\theta} \approx 0.01$, showing overdispersion as expected.

To determine which cells should be perturbed, we regressed the boolean assignment indicator vector as obtained from SCEPTRRE on the same covariates (now only `grna_n_nonzero` was significant).

Pick a desired total number of cells n . Let S_P be the indices of the cells SCEPTRRE concluded were perturbed, and let S_N be $n - |S_P|$ random indices from among the cells that SCEPTRRE did not consider perturbed. $S = S_P \cup S_N$ is our set of indices that we sample for, and assume it's sorted so the order is fixed.

Let Z be the $n \times p$ covariate matrix. Let μ be the fitted values from the NB regression on the UMI counts, and let $\hat{\varphi}$ be the fitted values (probabilities) from the logistic regression. Sample perturbation indicators $X_i \stackrel{iid}{\sim} \text{Ber}(\hat{\varphi}_i)$, while adjusting the logistic regression intercept (on the link scale) by φ_0 to control the overall perturbation rate.

Then sample from the NB mixture

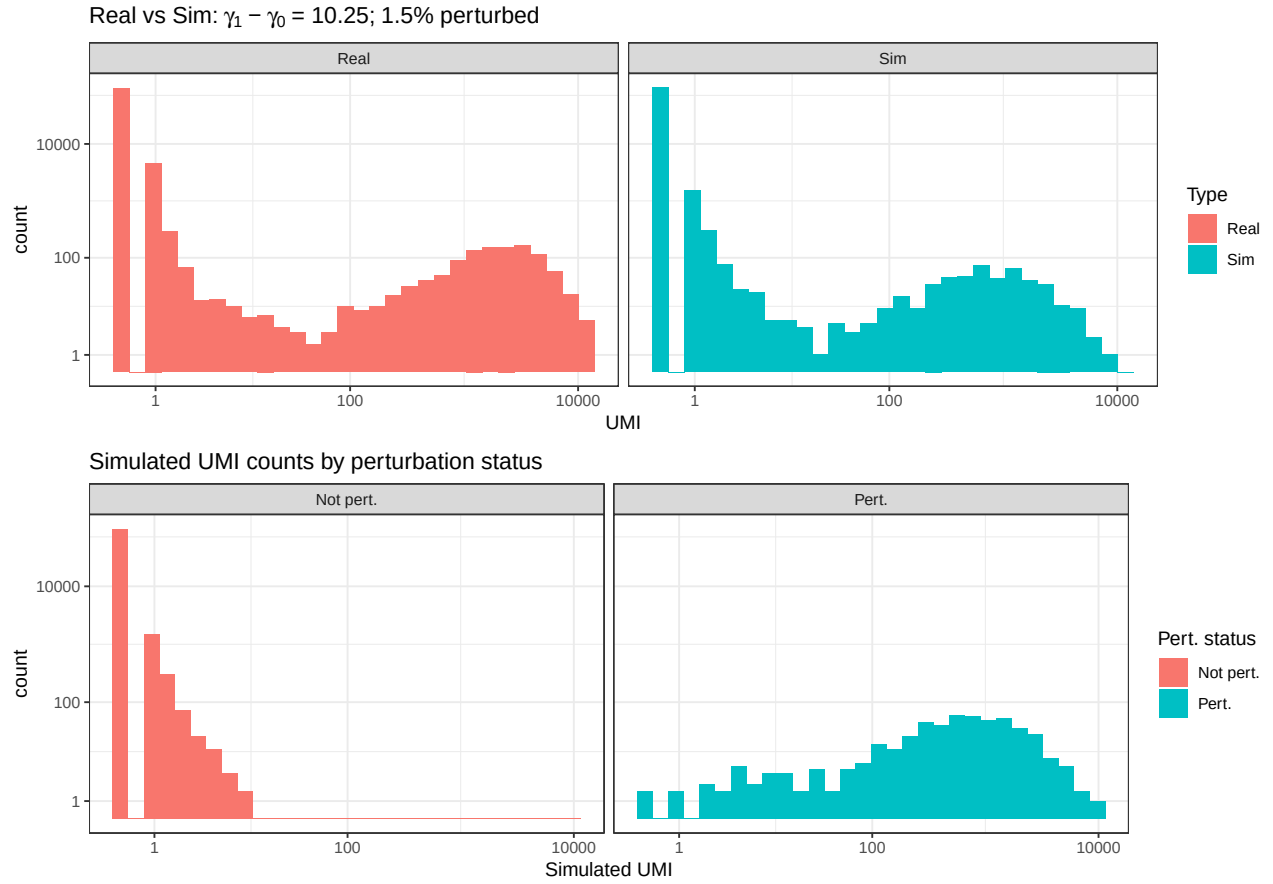
$$(1 - X_i)\text{NB}(\mu_i e^{\gamma_0}, \theta_0) + X_i \text{NB}(\mu_i e^{\gamma_1}, \theta_1)$$

so we have 5 parameters that control the simulation: the perturbation rate adjustment φ_0 ; the overdispersions for each component θ_0 and θ_1 ; and the respective mean adjustments γ_0 and γ_1 .

We picked these parameters approximately match on the fraction of 0 UMI counts; the mean; the variance; the maximum; and the overall shape.

We then ran each method on a simulated gRNA matrix with 25 guides each independently sampled from this process, and n cells for each. (TODO double-check that there is no information sharing between guides, like normalizations or something, which would make running iid guides as a single gRNA matrix a bad idea.)

SCEPTRRE was given the original gRNA matrix covariates rather than computing them from this simulation, since it doesn't make sense to compute them that way. (TODO or does it... ? yeah maybe this is still valid, and then less "cheating" for sceptre)



Method notes:

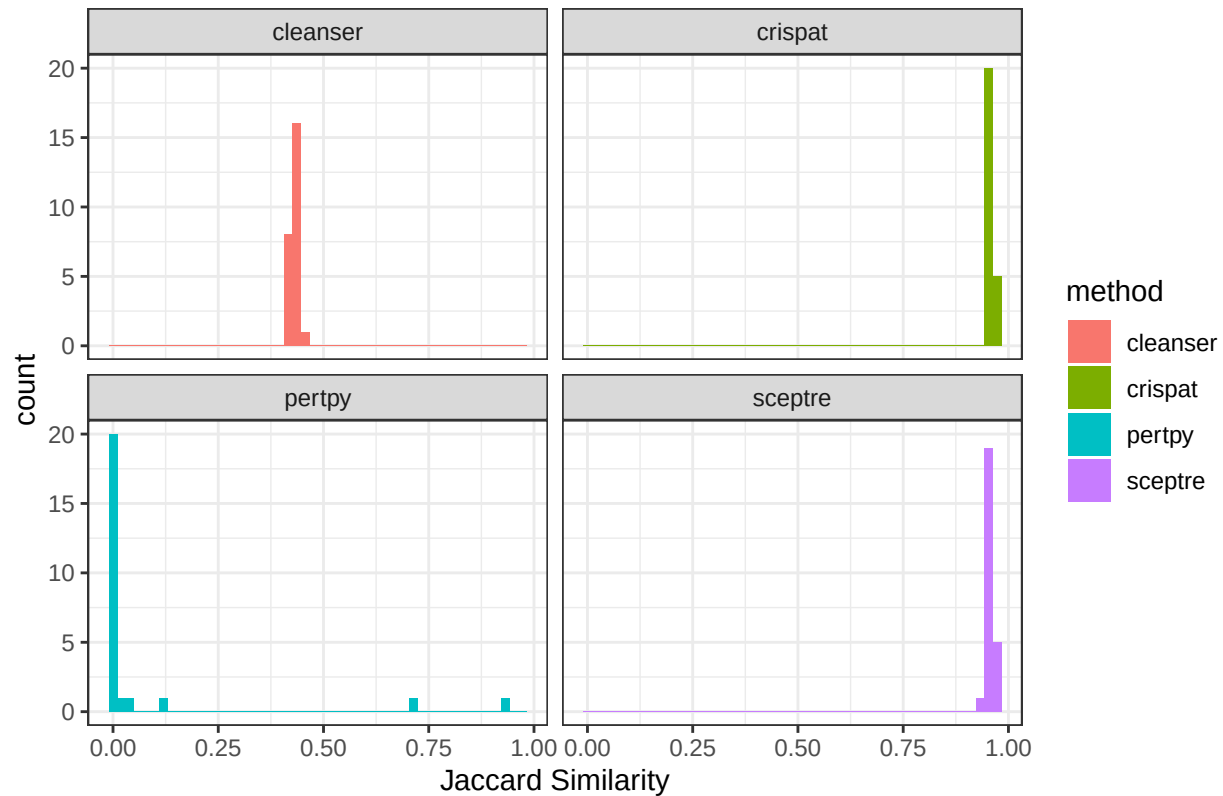
- CLEANSER had to be run separately on each piece, as it ran out of memory when all 25 simulated gRNAs were provided at once
- sceptre was allowed to control for `grna_n_nonzero` and `grna_n_umis`

2. Results

a. simulated data: comparison to ground truth

Here is one result: on a simulated dataset with 100k cells, and a boosted perturbation probability so 1.5% of cells are perturbed, we get the following. (TODO sceptre was run low MOI for these)

Similarities by method for replogle-rd7_simulated_100k_0.015-pert

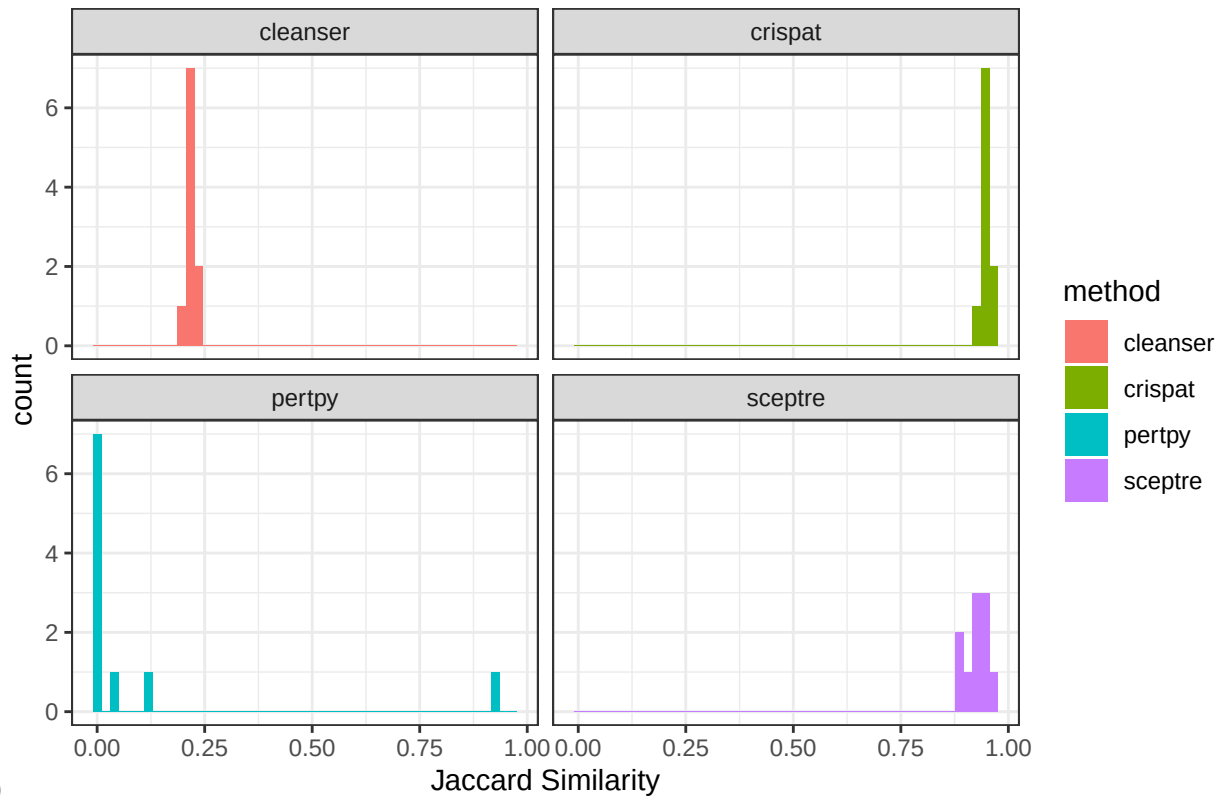


Observations:

- cleanser picks up on some signal but only partially gets it right
- crispat and sceptre do comparably well
- perpty also totally misses this one

now still 100k cells, but the perturbation rate is dropped. The only difference between these two is the intercept adjustment parameter being lowered, changing the overall perturbation rate. (TODO sceptre was run low MOI

Similarities by method for replegle-rd7_simulated_100k_0.005-pert



for these)

```
## $scep
## [1] 0.8878049 0.8942652 0.9077758 0.9268707 0.9269663 0.9315789 0.9413854
## [8] 0.9450549 0.9541833 0.9663551
##
## $crisp
## [1] 0.9303202 0.9463602 0.9468085 0.9470588 0.9472693 0.9502573 0.9519573
## [8] 0.9521277 0.9631676 0.9644860
##
## $pert
## [1] 0.000000000 0.000000000 0.000000000 0.000000000 0.000000000 0.000000000
## [7] 0.001702128 0.034023669 0.111111111 0.935201401
##
## $clns
## [1] 0.2041152 0.2081281 0.2106969 0.2137622 0.2152373 0.2184774 0.2189349
## [8] 0.2200247 0.2277433 0.2298665
```

another experiment: should sceptre be run on high MOI for these experiments...? that does seem right since each row is independent, so a cell's assignment in one place shouldn't affect it in another! (TODO sceptre was high low MOI for these)

```
rm(list=setdiff(ls(), c(objs_to_not_rm,
    "process_sceptre", "process_pert", "process_crispat", "process_clns", "process_clns_parts", "jaccard_similarity")),
    envir=globalenv())

dataset_names <- c("replegle-rd7_simulated_100k_0.005-pert", "replegle-rd7_simulated_100k_0.015-pert")
run_name <- "run_sceptre_all_sims"

output_base_fp <- file.path(
    .get_config_path("LOCAL_BENCHMARKING_DIR"),
```

```

"guide_assignment/outputs",
run_name
)

sims <- list()
for(dataset_name in dataset_names) {
  input_base_fp <- file.path(
    .get_config_path("LOCAL_BENCHMARKING_DIR"),
    "guide_assignment/input_data",
    dataset_name
  )

  ground_truth <- readRDS(file.path(input_base_fp, "true_perturbation_status.rds"))
  assn_list <- vector("list", length = nrow(ground_truth)) |>
    setNames(rownames(ground_truth))

  true_assign <- assn_list
  for(grna_name in names(true_assign)) {
    true_assign[[grna_name]] <- which(ground_truth[grna_name, ] == 1)
  }

  scep_assns <- process_sceptre(paste0(output_base_fp, "/assignment_matrix_sceptre_", dataset_name, ".rds"))

  sims[[dataset_name]] <- Map(jaccard, scep_assns, true_assign) |> as.numeric()
}

lapply(sims, sort)

## $`reprogle-rd7_simulated_100k_0.005-pert`
## [1] 0.8878049 0.8942652 0.9077758 0.9268707 0.9269663 0.9315789 0.9413854
## [8] 0.9450549 0.9541833 0.9663551
##
## $`reprogle-rd7_simulated_100k_0.015-pert`
## [1] 0.9308814 0.9456376 0.9458556 0.9491758 0.9492063 0.9532398 0.9544271
## [8] 0.9551074 0.9579376 0.9580878 0.9588235 0.9597315 0.9601100 0.9607438
## [15] 0.9611381 0.9621110 0.9623546 0.9625341 0.9626048 0.9628331 0.9649351
## [22] 0.9654472 0.9659546 0.9682003 0.9692308

```

makes no difference, so the low MOI results seem still valid. The SCEPTRE handbook also makes no reference to MOI in the assignment section.

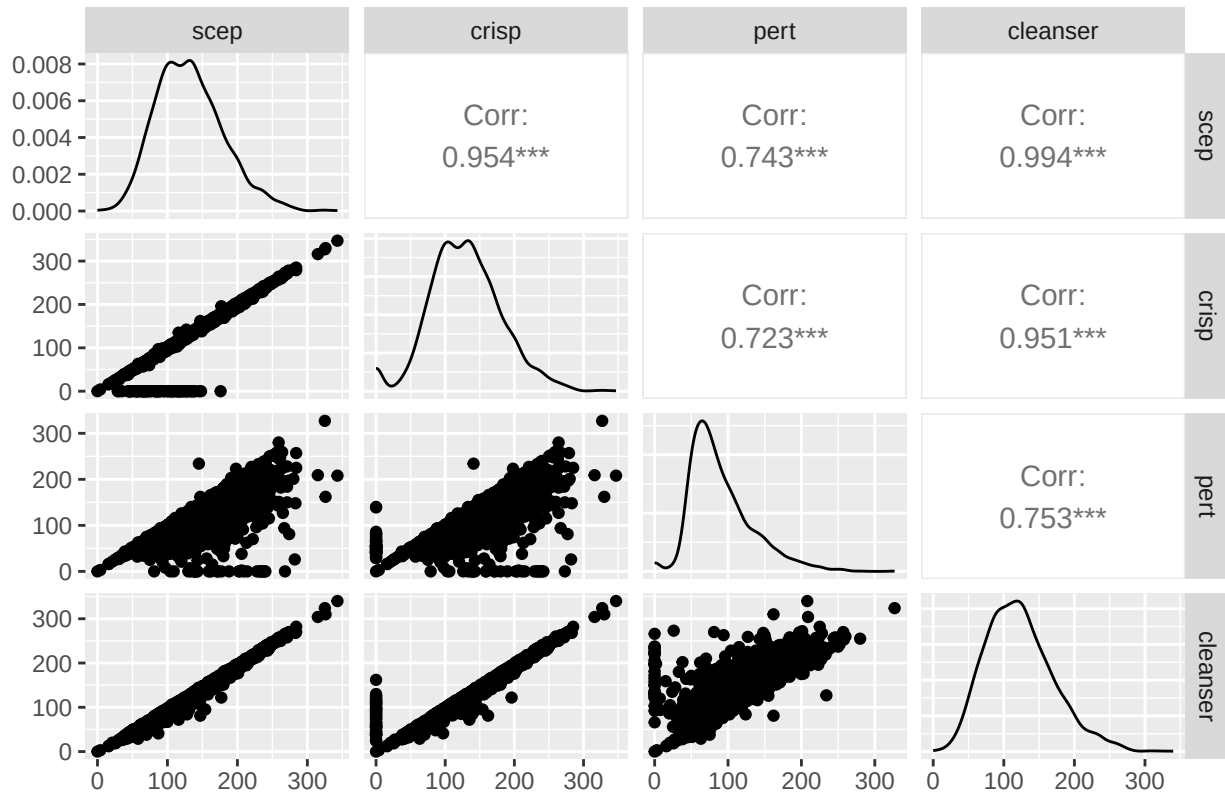
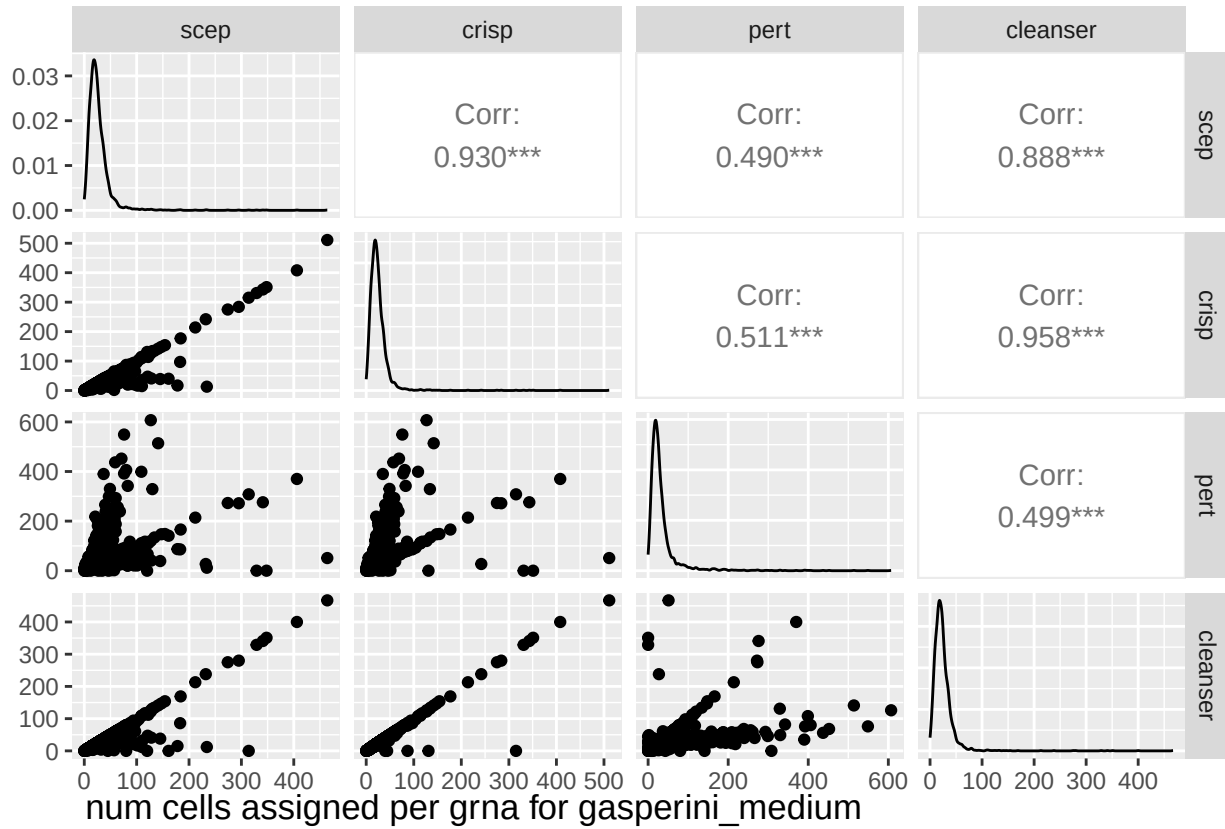
computational results for simulations



2. real data

this is all on the MEDIUM datasets due to how long it was taking (for pertpy and cleanser), and the memory (for cleanser).

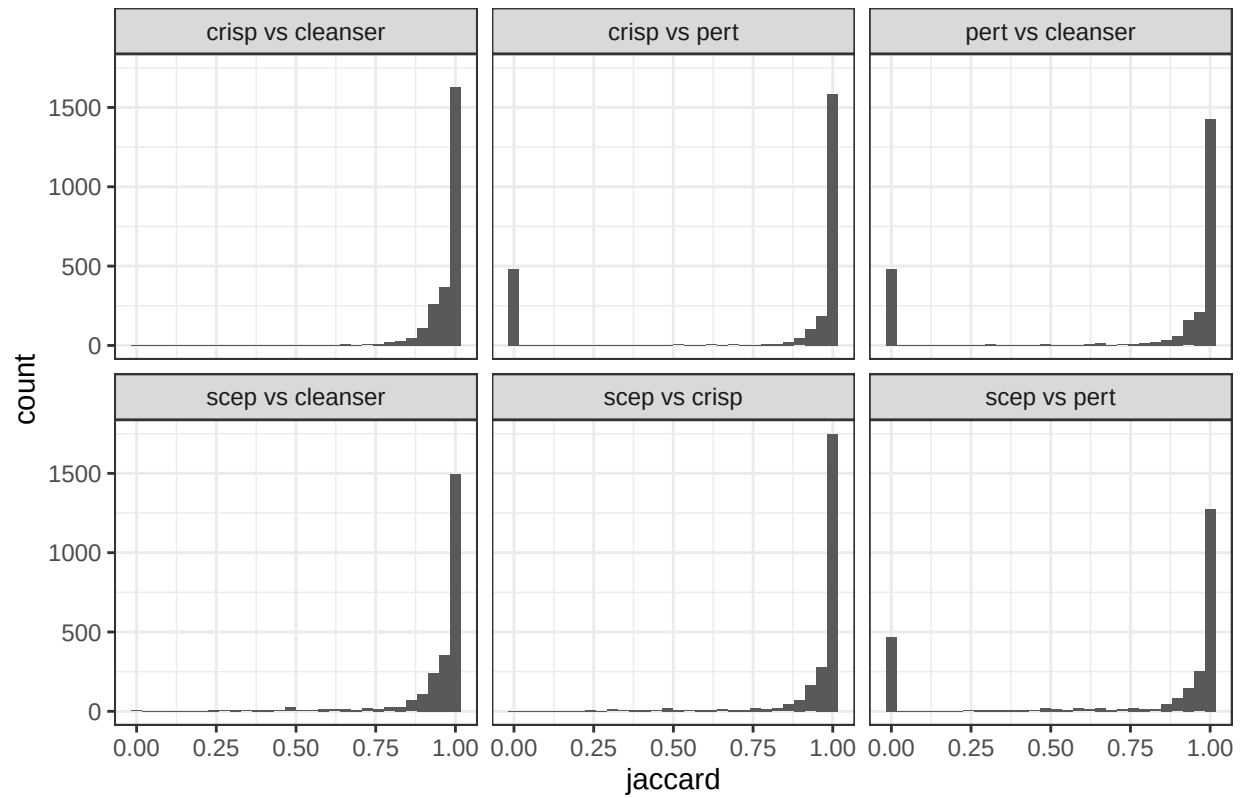
num cells assigned per grna for replogle-rd7_medium



Warning: Removed 13 rows containing non-finite outside the scale range


```
## (`stat_bin()`).
```

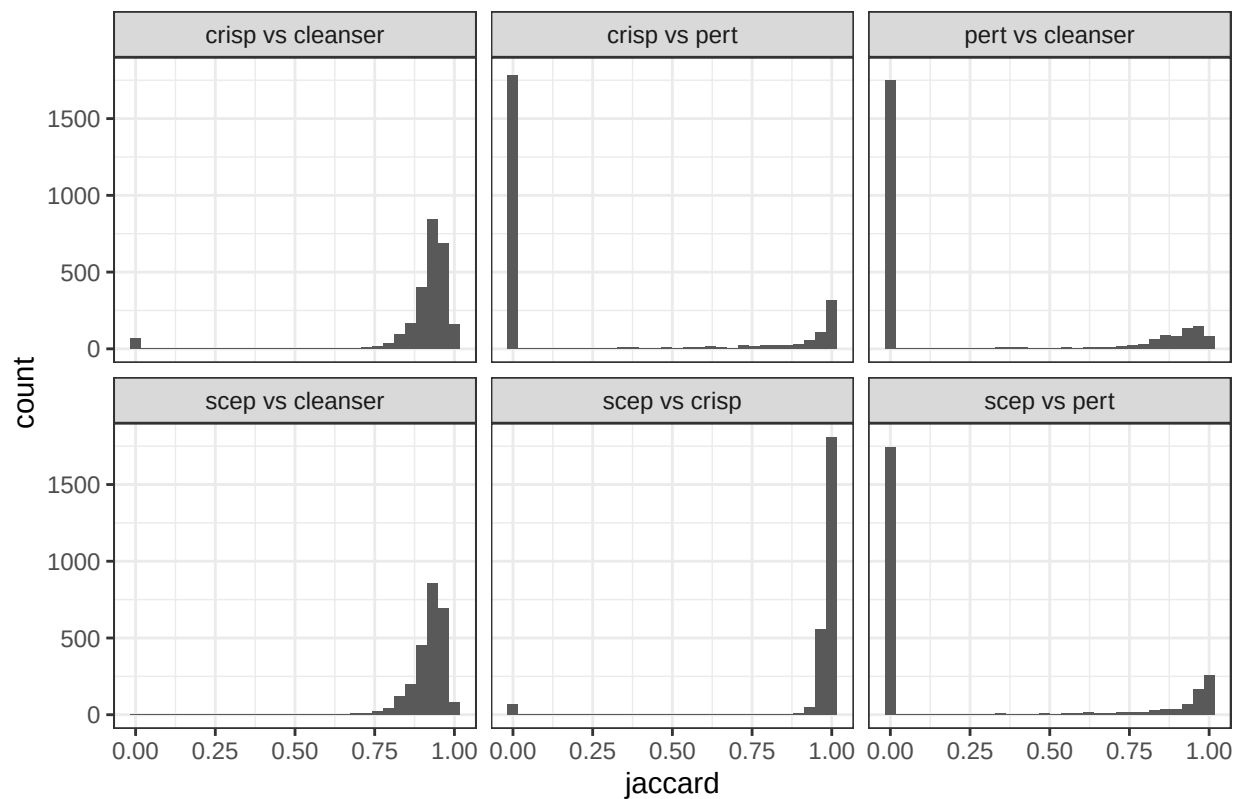
grna assignment similarities for replogle-rd7_medium



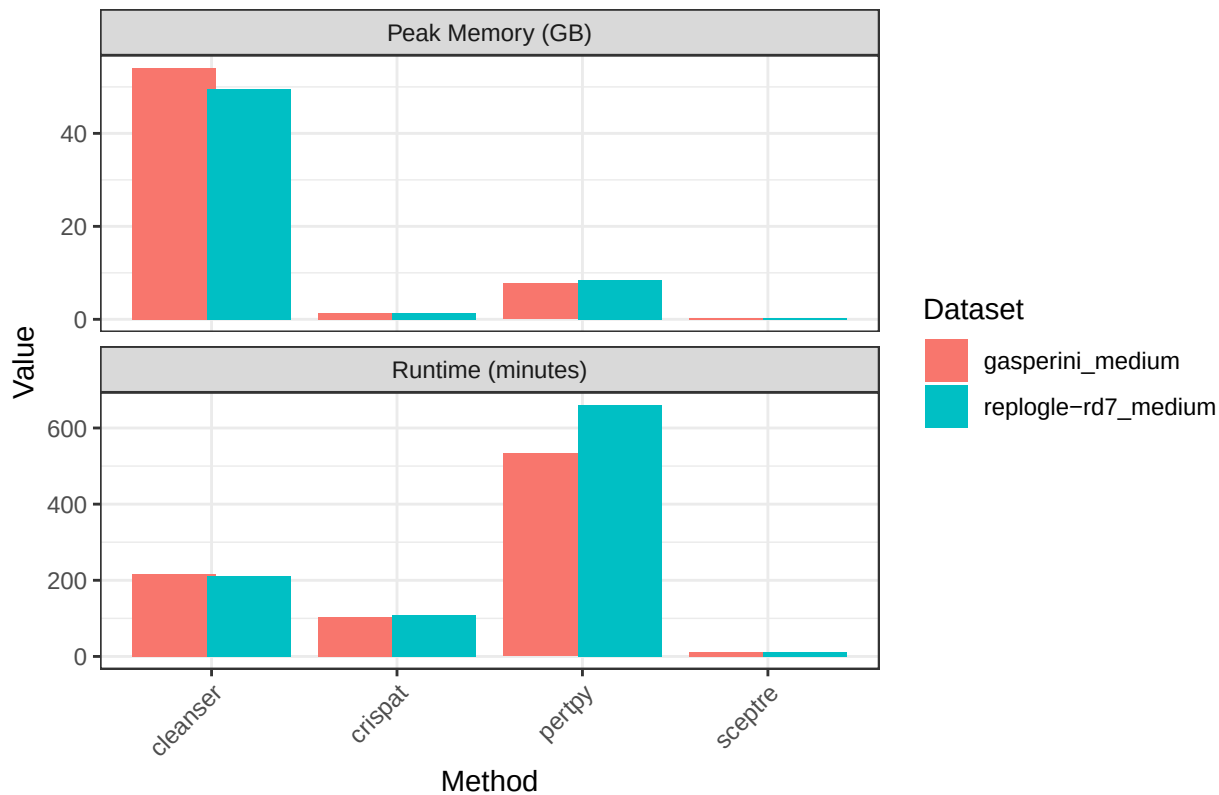
```
## Warning: Removed 6 rows containing non-finite outside the scale range
```

```
## (`stat_bin()`).
```

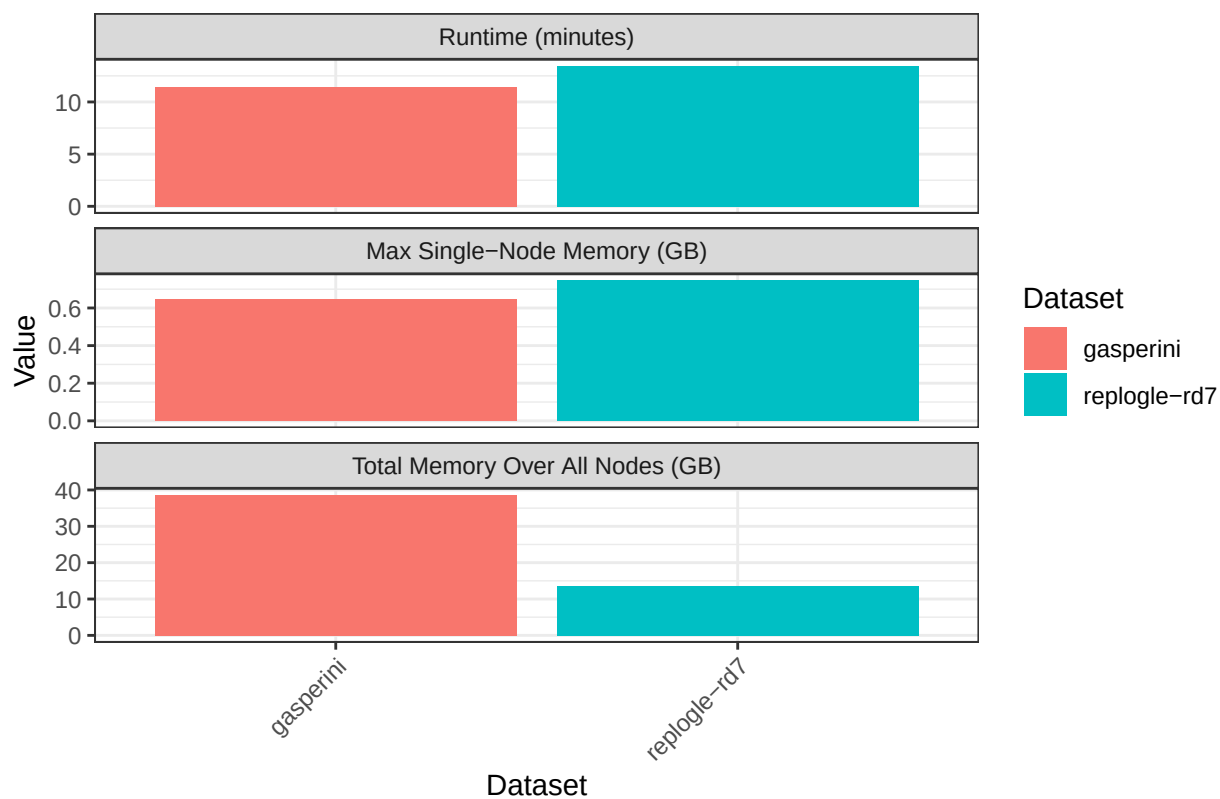
grna assignment similarities for gasperini_medium



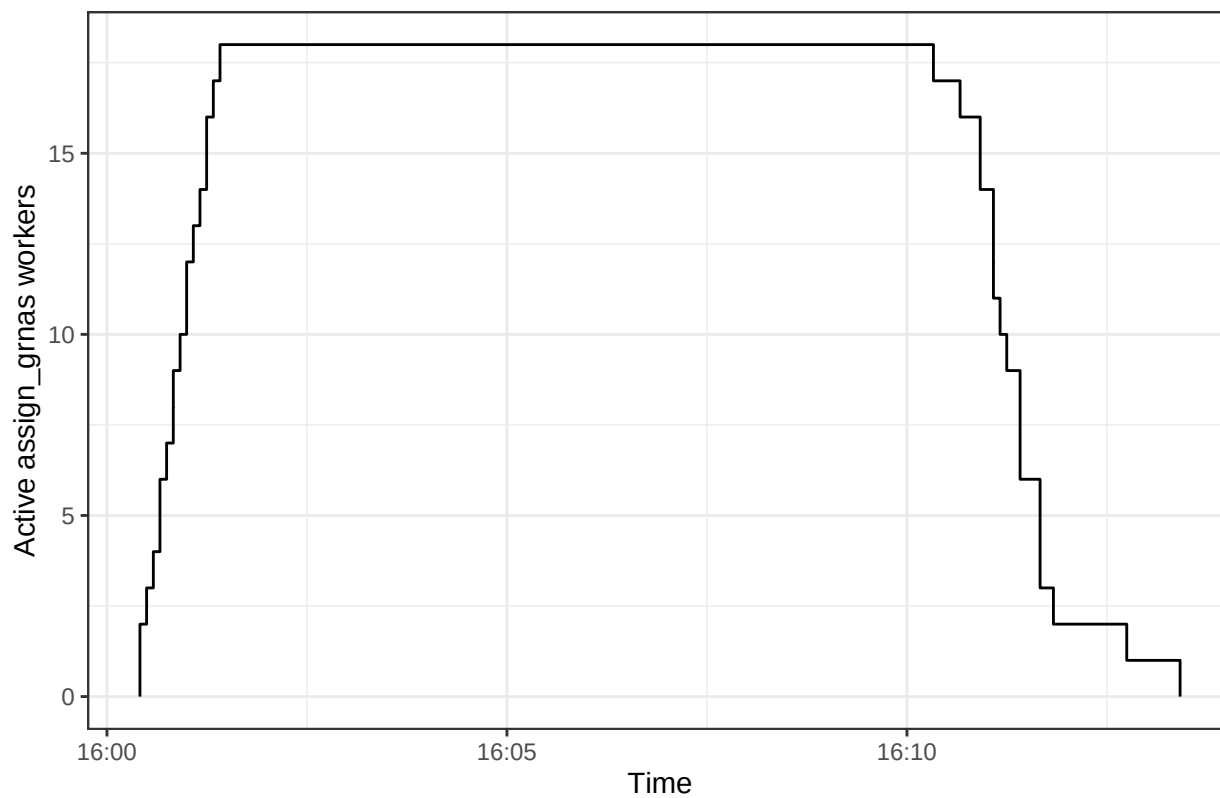
Computational Performance by Method and Dataset

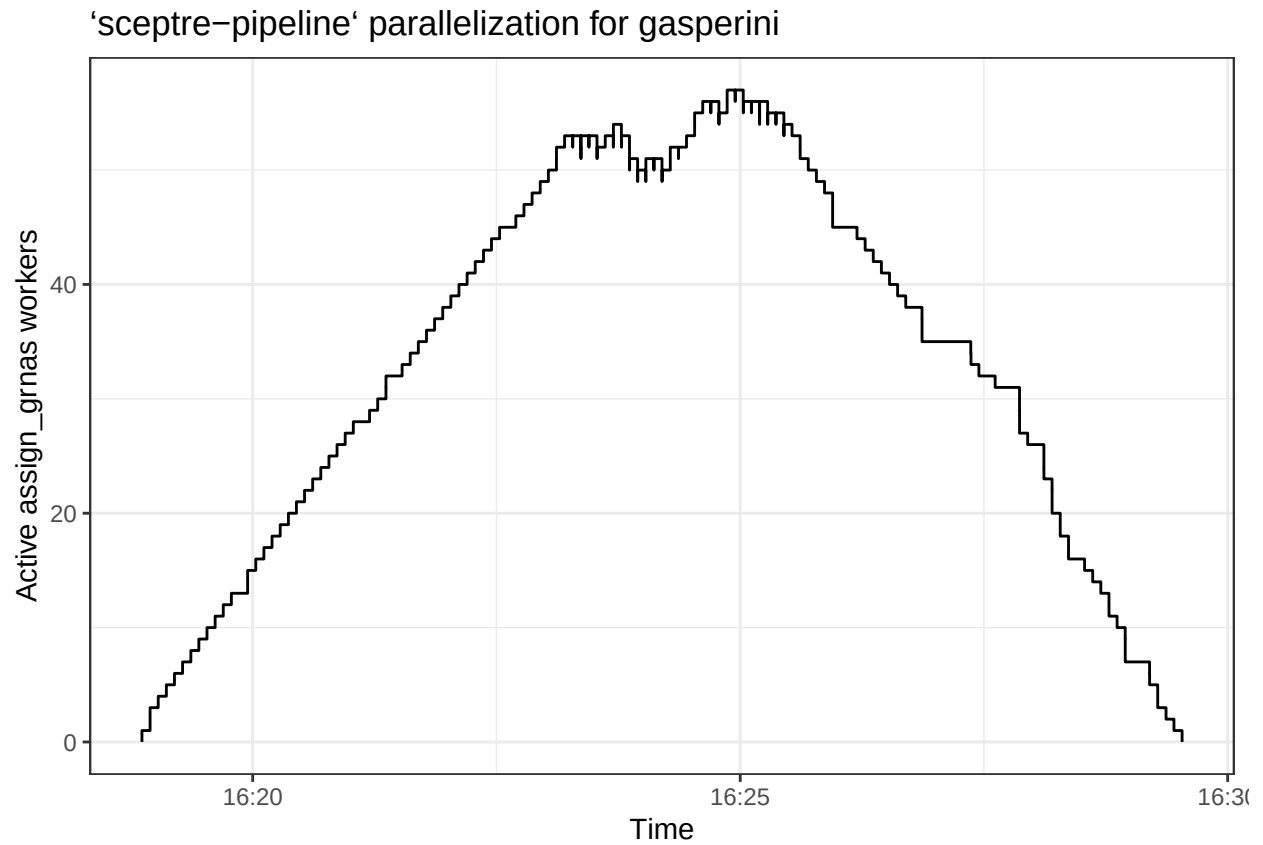


Sceptre-Pipeline Performance by Dataset



'sceptre-pipeline' parallelization for replogle-rd7





TODO: array jobs to make these more overlapping?

NOTES FROM fishash PAPER

- barnyard dataset w/ partial ground truth
- different performance for sceptre in theirs vs ours (and other methods too)
- fancier simulation
- different metric for agreement
- add more methods – geomux?